

WHATSNEXT VISION MOTORS

Phase 5 Comprehensive Documentation: Deployment, Documentation & Maintenance

Project Track: Salesforce CRM Enterprise Solution (Academic & Professional Specification)

Document Focus: Phase 5 - Deployment Strategy, System Maintenance, Documentation & Demo Video Guidelines

Target Environment: Developer Org (Simulated Production Migration)

Document Status: Final Technical Deliverable & Operations Manual

1. Executive Summary & Project Overview

WhatsNext Vision Motors is a Salesforce-based CRM solution that modernizes vehicle ordering and dealership operations. The platform automates vehicle order placement, validates stock availability, assigns each order to the nearest dealer based on customer location, updates order statuses dynamically, and sends scheduled email reminders for test drives. It replaces manual registers, spreadsheets and disconnected systems with one centralized, automated platform.

1.1 High-Level Business Objectives

- **Operational Efficiency:** Unify vehicles, dealers, customers, orders, test drives and service requests in a single Salesforce instance.
- **Process Automation:** Use Record-Triggered Flows, Apex triggers, Batch Apex and Scheduled Apex to assign dealers, block out-of-stock orders, update stock and send test drive reminders.
- **Data Governance:** Enforce role-based access control so that administrators, sales coordinators and inventory managers see only what they need.
- **Analytics & Insights:** Provide reports and dashboards on order status, dealer performance and sales metrics for real-time decision making.
- **Customer Satisfaction:** Give customers accurate availability, faster order processing and timely test drive reminders.

1.2 Problem Statements Addressed

ID	Persona	Problem	Impact
PS-1	Operations Manager	Stock information is manual and systems are disconnected	Stressed; wrong orders, delays and cancellations
PS-2	Customer	Vehicle availability is unclear and updates are delayed	Uncertain; poor ordering experience

2. Summary of Implementation Phases (Phases 1-4)

Before Phase 5, WhatsNext Vision Motors went through ideation, requirement analysis, solution design, agile planning and development.

Phase	Core Focus Area	Key Deliverables & Artifacts
Phase 1	Ideation & Requirement Analysis	Problem statements, empathy map (Operations Manager), idea prioritization (Salesforce CRM selected for automation, scalability and real-time visibility), customer journey map, functional and non-functional requirements, data flow and technology stack.
Phase 2	Project Design & Planning	Problem-solution fit, proposed solution, solution architecture, and agile plan (Epics > Stories > Story Points) across six sprints with velocity tracking.
Phase 3	Data Modeling & UI	Custom objects (Vehicle__c, Vehicle_Dealer__c, Vehicle_Customer__c, Vehicle_Order__c, Vehicle_Test_Drive__c, Vehicle_Service_Request__c), custom tabs, fields, picklists, lookup relationships and the 'WhatNext Vision Motors' Lightning App.
Phase 4	Automation & Apex Development	'Auto Assign Dealer' and 'Test Drive Reminder' Record-Triggered Flows, VehicleOrderTrigger with VehicleOrderTriggerHandler (stock validation), VehicleOrderBatch (pending order processing) and Scheduled Apex.

2.1 Functional Requirements Coverage

FR No	Requirement	Delivered By
FR-1	Vehicle management	Vehicle__c object, tab, Stock_Quantity__c and Status__c fields
FR-2	Dealer management	Vehicle_Dealer__c object with location, phone and email
FR-3	Customer order creation	Vehicle_Order__c linked to Vehicle_Customer__c and Vehicle__c
FR-4	Stock validation	VehicleOrderTrigger / preventOrderIfOutOfStock()
FR-5	Dealer auto-assignment	Auto Assign Dealer record-triggered flow
FR-6	Test drive scheduling	Vehicle_Test_Drive__c and Test Drive Reminder flow
FR-7	Reports & dashboards	Order, dealer and sales reports and dashboards

3. Phase 5: Deployment Strategy & DevOps Simulation

All configurations, customizations and automations for WhatsNext Vision Motors were designed, implemented and validated inside a Salesforce Developer Edition Org. This section describes the enterprise deployment methodology simulated for production readiness.

3.1 Environment Rationale: Developer Org vs. Enterprise Pipeline

- **Standalone Environment Architecture:** Developer Edition orgs are isolated and have no direct outbound change-set connection to a live production org.
- **Enterprise CI/CD Requirements:** Enterprise rollouts use a multi-tier pipeline: Developer Sandbox > Partial Copy Sandbox > Full Copy Sandbox > Production, governed by continuous integration tooling.

3.2 Enterprise Migration & Metadata Management Strategy

To prepare the solution for commercial migration, all components are organized into metadata packages:

Metadata Category	Salesforce Metadata API Types	WhatsNext Artifacts Included
Data Architecture	CustomObject, CustomField, CustomTab	Vehicle__c, Vehicle_Dealer__c, Vehicle_Customer__c, Vehicle_Order__c, Vehicle_Test_Drive__c, Vehicle_Service_Request__c and their lookup fields.
Business Logic	ApexClass, ApexTrigger, Flow	VehicleOrderTriggerHandler, VehicleOrderTrigger, VehicleOrderBatch, Auto Assign Dealer flow, Test Drive Reminder flow.
User Interface	CustomApplication, CustomTab	WhatNext Vision Motors Lightning App and object tabs.
Security & Access	Profile, PermissionSet	System Administrator profile access to the app; role-based access for coordinators and inventory managers.
Analytics	Report, Dashboard	Order Status, Dealer Performance and Sales Metrics reports and dashboards.

3.3 Deployment Tooling & CI/CD Pipeline

- **Source Code Versioning:** Metadata is retrieved with SFDX CLI (sf project retrieve start) and committed to a Git repository.
- **Automated Testing & Validation:** Continuous Integration runs Apex unit tests (sf apex run test) whenever code is merged to the integration branch.
- **Change Sets & DevOps Center:** Declarative components (flows, objects, apps) are moved with Change Sets or DevOps Center, which track dependencies to avoid missing-component failures.
- **Deployment Order:** Objects and fields > Apex classes and triggers > Flows > Tabs and App > Reports and Dashboards > Permissions.

3.4 Pre-Deployment Checklist

- All Apex classes and triggers have test classes with at least 75% coverage (target 85%).
- Both flows are tested and activated; the Test Drive flow scheduled path verified.
- Batch Apex tested with a pending order and restocked vehicle; Scheduled Apex job configured.
- Page layouts, tabs and Lightning App assigned to the correct profiles.
- Reports and dashboard folders shared with the right users.

4. Maintenance, Monitoring & Troubleshooting

Post-deployment stability requires continuous monitoring, proactive data hygiene and structured troubleshooting.

4.1 Ongoing Maintenance Activities

- **Data Quality & Hygiene:** Weekly review of duplicate customer and dealer records, correct Dealer_Location__c and customer Address values (they drive dealer assignment), and periodic archival of delivered or canceled orders.
- **Inventory Accuracy:** Regular reconciliation of Stock_Quantity__c against physical stock; verify Vehicle Status__c (Available, Out of Stock, Discontinued).
- **User Offboarding & Access Audits:** Quarterly audits of active licenses, profile assignments and permission sets.

- **Salesforce Release Readiness:** Review Spring, Summer and Winter releases and test flows and Apex for backward compatibility.

4.2 System Monitoring Framework

- **Apex Exception Logs:** Monitor the Developer Console and Setup debug logs for uncaught exceptions in the order trigger and handler.
- **Flow Engine Monitoring:** Review Paused and Failed Flow Executions for the Auto Assign Dealer and Test Drive Reminder flows.
- **Batch & Scheduled Job Tracking:** Audit Apex Jobs and Scheduled Jobs (AsyncApexJob) to confirm the order batch runs and completes without queue limits.
- **Email Deliverability:** Confirm test drive reminder emails are sent and not blocked by deliverability settings.

4.3 Troubleshooting & Incident Resolution Guide

Issue Category	Common Root Cause	Diagnostic Tool / Area	Resolution & Corrective Action
Order Blocked	Vehicle Stock_Quantity__c is 0 or less (intended behaviour)	Vehicle record; trigger error message	Add stock to the vehicle; the pending order is then confirmed by the batch job.
Dealer Not Assigned	Customer Address does not exactly match any Dealer_Location__c	Flow debug; Get Nearest Dealer element	Standardize location values or add a fault/default path in the flow.
Flow Execution Fault	Null value passed into a required field	Paused & Failed Flow Executions	Add Fault Path and null checks.
Reminder Email Not Sent	Test drive status not 'Scheduled' or scheduled path offset misconfigured	Flow entry conditions; scheduled path	Verify Status__c = Scheduled and Test_Drive_Date__c; reactivate flow.
Batch Not Updating Orders	Compile error (e.g., field API names typed incorrectly) or job not scheduled	Apex Jobs; Developer Console	Correct API names (Status__c, Vehicle__c, Vehicle_Order__c) and reschedule the job.
Apex CPU Time Limit	SOQL or DML inside loops	Debug Logs / Developer Console	Bulkify code using Sets and Maps and a single DML per collection.
User Access Denied	Missing profile or permission set	User Detail > Permission Set Assignments	Grant the required access or adjust sharing.

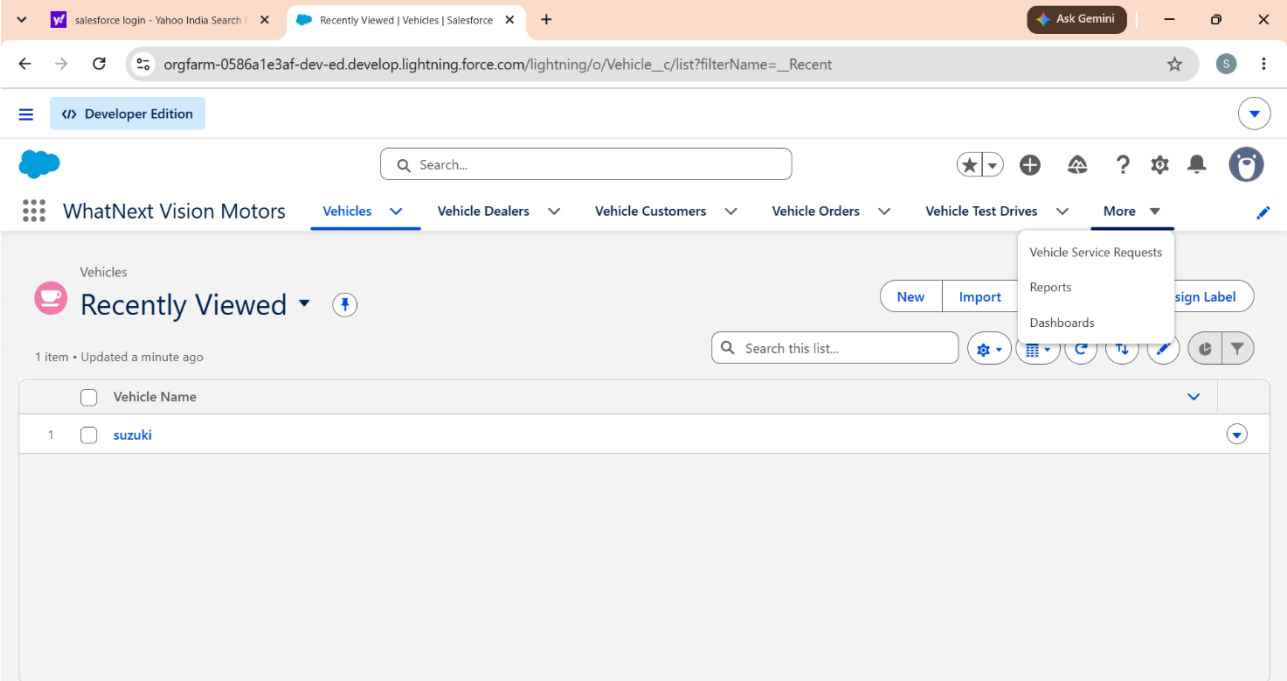
5. Project Documentation & Feature Evidence Placeholder Specification

5.1 Core Artifact Inventory

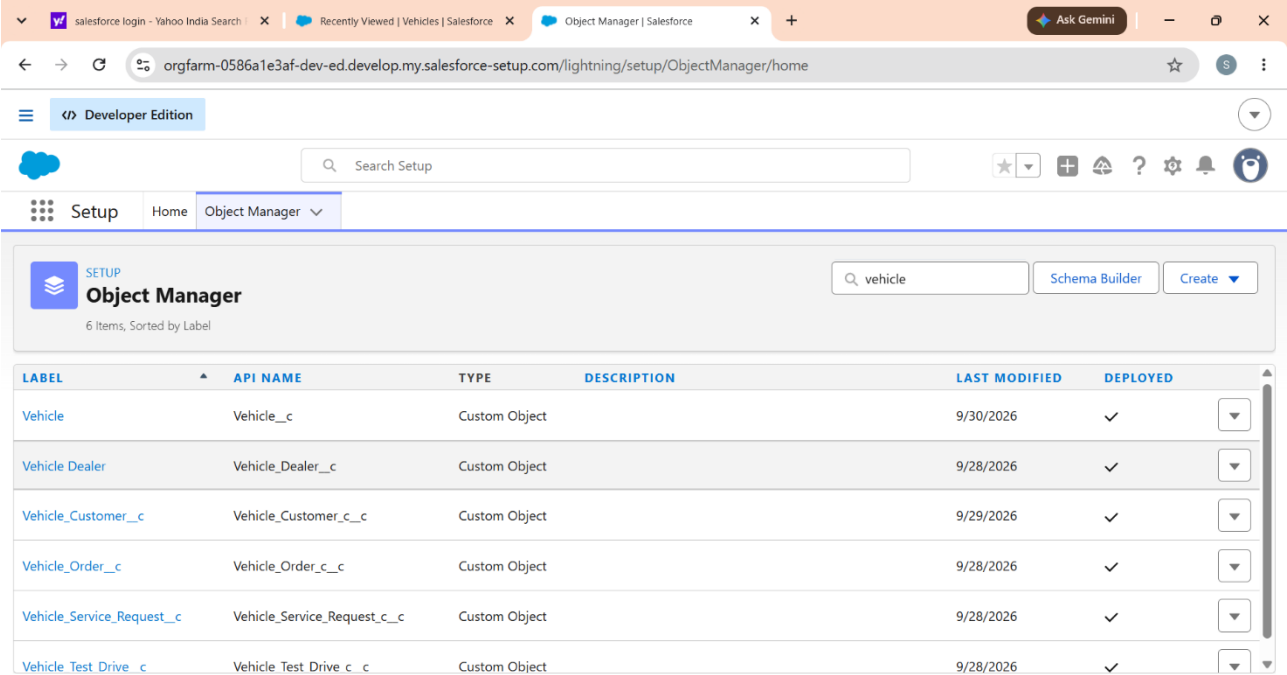
- **Data Architecture:** Business requirements and data model (ER diagram of Vehicle, Dealer, Customer, Order, Test Drive and Service Request objects).
- **Automation Architecture:** Flow diagrams, Apex trigger and handler logic, batch job design.
- **Planning Records:** Product backlog, sprint schedule, velocity and burndown chart.
- **Security Governance:** Profile and permission matrix.

5.2 System Visual Evidence Placeholders

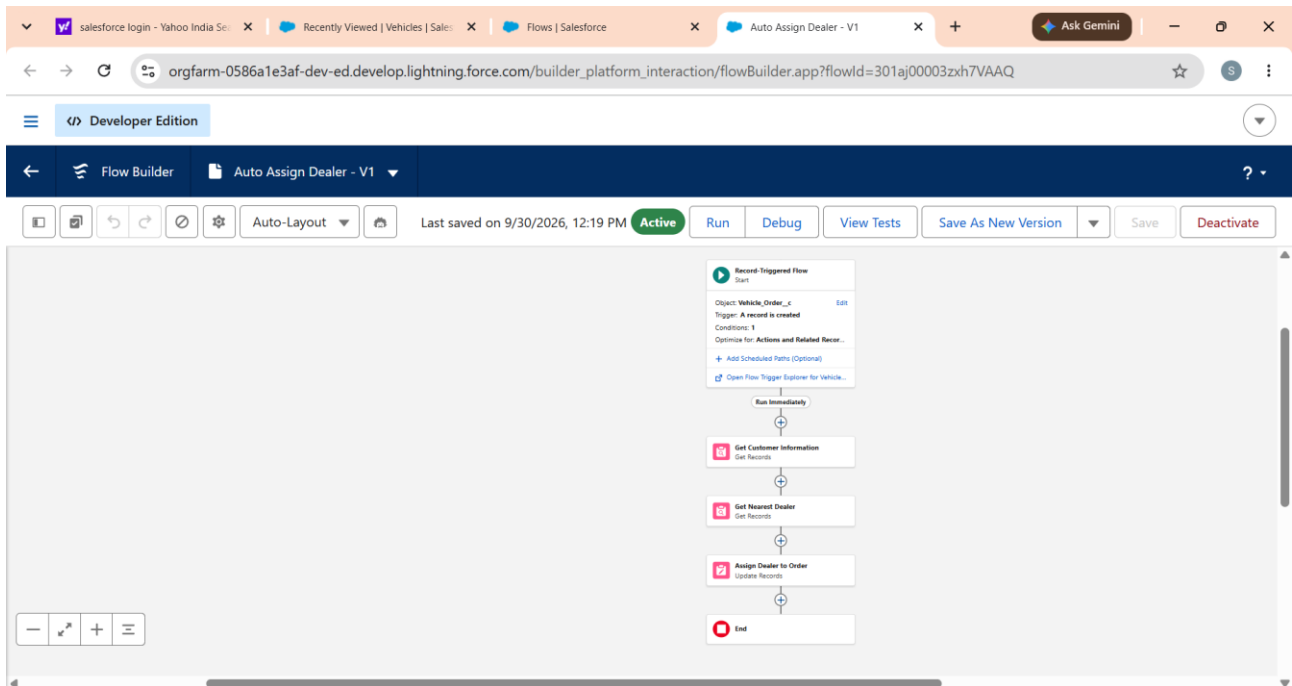
Screenshot 5.1: WhatNext Vision Motors Lightning App & Navigation - Custom app with Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard tabs.



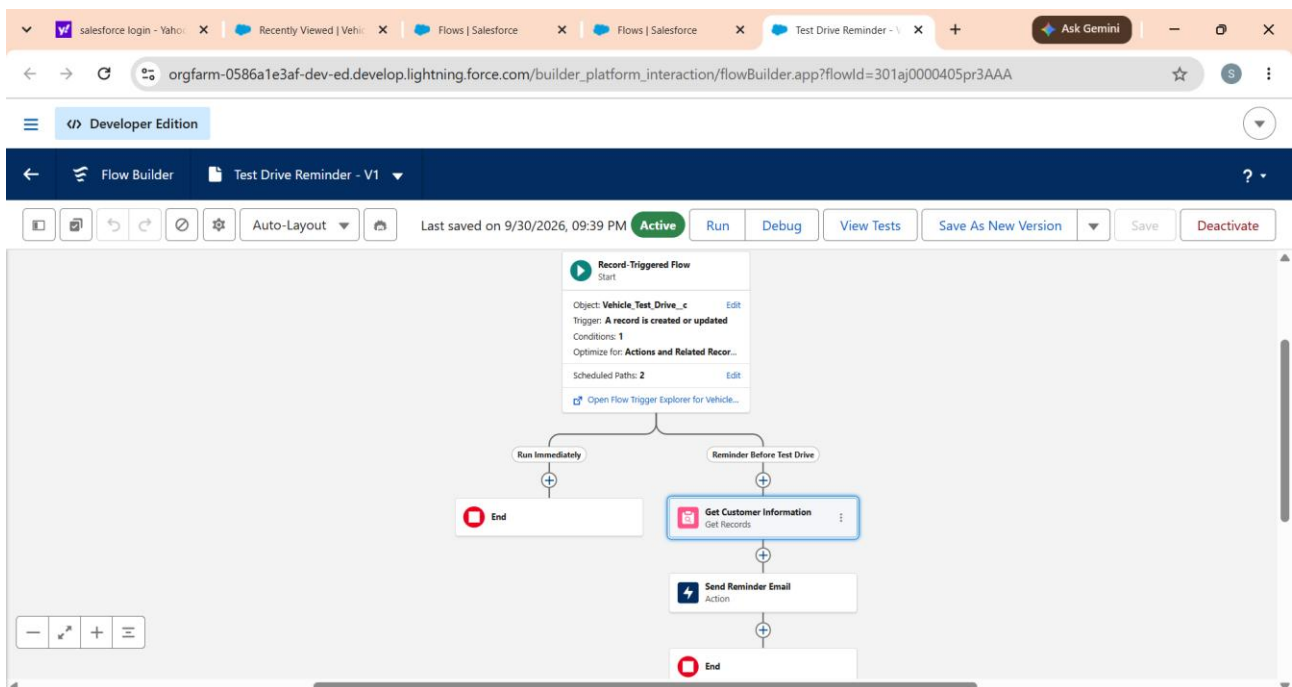
Screenshot 5.2: Object Manager Schema - Custom fields, picklists (Vehicle Model, Status) and lookup relationships.



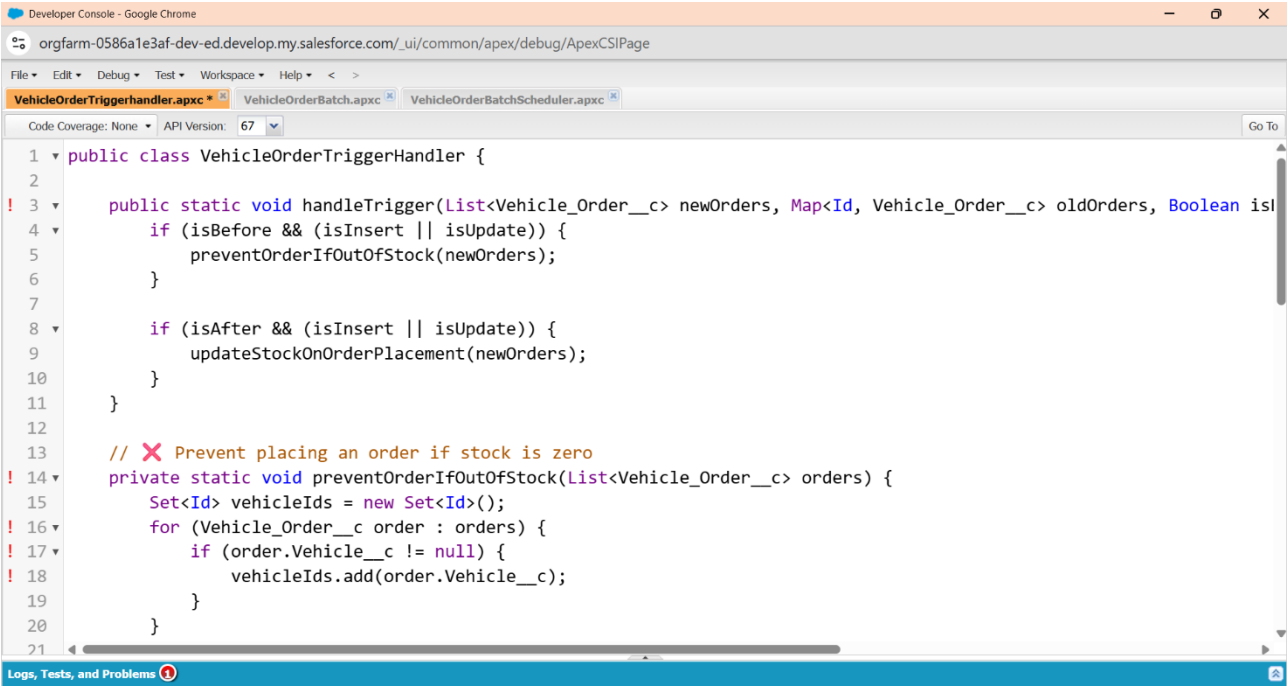
Screenshot 5.3: Auto Assign Dealer Flow- Flow canvas with Get Customer Information, Get Nearest Dealer and Assign Dealer to Order elements.



Screenshot 5.4: Test Drive Reminder Flow - Scheduled path (1 day before), Get Records and Send Email action.



Screenshot 5.5: Apex Classes, Trigger and Batch Job - VehicleOrderTriggerHandler, VehicleOrderTrigger and VehicleOrderBatch in Developer Console.



```
1 public class VehicleOrderTriggerHandler {
2
3     public static void handleTrigger(List<Vehicle_Order__c> newOrders, Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore, Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
4         if (isBefore && (isInsert || isUpdate)) {
5             preventOrderIfOutOfStock(newOrders);
6         }
7
8         if (isAfter && (isInsert || isUpdate)) {
9             updateStockOnOrderPlacement(newOrders);
10        }
11    }
12
13    // X Prevent placing an order if stock is zero
14    private static void preventOrderIfOutOfStock(List<Vehicle_Order__c> orders) {
15        Set<Id> vehicleIds = new Set<Id>();
16        for (Vehicle_Order__c order : orders) {
17            if (order.Vehicle__c != null) {
18                vehicleIds.add(order.Vehicle__c);
19            }
20        }
21    }
22 }
```

6. Guidelines & Structural Script for Project Demo Video

The demo video walks stakeholders through the WhatsNext Vision Motors system. Follow the 8-step script below.

6.1 Step-by-Step Presentation Script (Target Duration: 5-10 Minutes)

Step	Section Title	Target Time	Demonstration Script & Key Visual Focus
1	Introduction	0:00 - 0:45	Introduce yourself, the project title and the business goals.
2	App Overview	0:45 - 1:30	Open the App Launcher, select WhatNext Vision Motors and show the navigation items.
3	Data Model & UI	1:30 - 3:00	Show objects, fields and relationships; create a vehicle, dealer and customer record.
4	Order & Stock Validation	3:00 - 4:30	Place an order for an in-stock vehicle, then for an out-of-stock vehicle to show the Apex validation error.
5	Automation Showcase	4:30 - 6:30	Show the Auto Assign Dealer flow, the Test Drive Reminder flow, the batch job and scheduled Apex.
6	Reports & Dashboards	6:30 - 7:45	Display order status, dealer performance and sales dashboards.
7	Debugging & Fixes	7:45 - 8:45	Discuss challenges, debug logs and how errors were resolved.
8	Conclusion	8:45 - 9:30	Summarize outcomes, learnings and thank evaluators.

6.2 Presentation Quality & Technical Pro Tips

- **Voice & Audio Quality:** Use clear audio with low background noise and a professional tone.

- **Display Resolution:** Record at 1080p with browser zoom of 100%-110%.
- **Logical Flow:** Intro > UI > Automation > Reports > Conclusion.
- **Highlight Key Automations:** Spend extra time on dealer auto-assignment, stock validation and the test drive reminder.

7. Conclusion & Project Outcomes

The WhatsNext Vision Motors CRM shows how Salesforce can streamline vehicle ordering through custom objects, record-triggered flows, Apex triggers, batch and scheduled jobs, and real-time reporting. It reduces manual intervention, improves order accuracy and raises customer satisfaction.

Key Outcomes & Technical Learnings

- **Data Modeling:** Designed objects, fields and lookup relationships for vehicle, dealer, customer and order data.
- **Automation Expertise:** Built Record-Triggered Flows and Apex (trigger, handler, Batch and Scheduled Apex).
- **UI Design:** Built a Lightning App with tabs for every module.
- **Lifecycle Management:** Covered the full lifecycle from ideation and agile planning to development and a production-migration simulation.

Although built in a Developer Edition Org for academic purposes, the project reflects real-world CRM best practices and is ready for future enterprise scaling.

Final Acknowledgment: Thank you for reviewing the WhatsNext Vision Motors Phase 5 Documentation!